

Taeyun Roh

Seoul, Republic of Korea | txnrd.github.io | github.com/txnrd

Researcher in AI systems and multimodal reasoning with experience spanning agentic memory, efficient LLM inference, healthcare AI, and applied product development.

Education

Korea University <i>M.S. in Computer Science and Engineering</i>	Mar. 2026 – Present Seoul, Republic of Korea
Korea University <i>B.S. in Computer Science and Engineering</i>	Mar. 2022 – Feb. 2026 Seoul, Republic of Korea

- Master’s student in Computer Science and Engineering.
- Undergraduate studies focused on systems, machine learning, and software engineering.

Research Experience

Data Mining & Information Systems Laboratory, Korea University <i>Undergraduate Researcher, advised by Prof. Jae Woo Kang</i>	Mar. 2025 – Present Seoul, Republic of Korea
Network and Mobile System Laboratory, KAIST <i>Research Intern, advised by Prof. Sung-Ju Lee</i>	Jan. 2025 – Mar. 2025 Daejeon, Republic of Korea
System Software Laboratory, Korea University <i>Undergraduate Researcher, advised by Prof. Gyeongsik Yang</i>	Jan. 2024 – Dec. 2024 Seoul, Republic of Korea

- Research on agentic memory systems for AI agents.
- Worked on efficient LLM inference, including on-device LLM inference.
- Researched federated learning for healthcare applications under resource heterogeneity.

Selected Publications

CLAG: Adaptive Memory Organization via Agent-Driven Clustering for Small Language Model Agents Taeyun Roh, Wonjune Jang, Junha Jung, Jaewoo Kang <i>ACL 2026 Findings</i>	2026 Published
Breaking Failure Cascades: Step-Aware Reinforcement Learning for Medical Multimodal Reasoning Junha Jung, Minbyul Jeong, Suhyeon Lim, Sungwook Jung, Jaehoon Yun, <u>Taeyun Roh</u> , Mujeen Sung, Jaewoo Kang <i>Preprint</i>	2026 Under Review
When Choices Become Priors: Contrastive Decoding for Scientific Figure Multiple-Choice QA <u>Taeyun Roh</u> , Eun-yeong Jo, Wonjune Jang, Jaewoo Kang <i>Preprint</i>	Preprint

Industry & Development Experience

Aigensciences <i>AI Researcher</i>	Mar. 2025 – Aug. 2025 Seoul, Republic of Korea
Blurting <i>Frontend Developer, Co-founder</i>	Jun. 2023 – Jan. 2025 Seoul, Republic of Korea

- Built an LLM-based pipeline for automated hypothesis generation in drug discovery.
- Developed a pediatric counseling chatbot designed for KakaoTalk integration.
- Co-founded a four-member startup and led end-to-end frontend development.
- Developed and operated a blind-dating application that reached #73 in the Korea App Store Social Networking category.

- Built mathematical models for optimizing teaching assistant compensation.
- Applied regression analysis to study the relationship between student study time and academic performance.

Awards

- **1st Place, Korea University Capstone Design** (Prize: KRW 1,800,000) — Developed *NEWSQRAB*, an AI-driven system for automated news-to-short-form video generation using RAG-based LLMs.
- **2nd Place, Korea University NEXT Software Competition** (Prize: KRW 1,000,000) — Recognized as lead AI engineer for the *NEWSQRAB* project.
- **3rd Place, 2024 KIISE Conference (Undergraduate Category)** (Dec. 2024) — Awarded for the paper “*Analysis on Research for Data and Resource Heterogeneity in Split-Federated Learning.*”
- **2nd Place, 2023 Korea University Hackathon** (Prize: KRW 1,000,000) — Built Noonchi, a real-time visitor prediction and recommendation service for Han River parks.

Selected Projects & Skills

Projects

- **Xalute** — Wearable health monitoring app developed with Korea University Anam Hospital for real-time health tracking and alerts.
- **Kakao Pediatric Response Chatbot** — LLM-based pediatric Q&A chatbot developed with Kakao Healthcare and Upstage.

Skills

- **Research:** Agentic memory, multimodal reasoning, LLM inference, healthcare AI, federated learning
- **Development:** Frontend development, applied LLM products, data analysis
- **Language:** English (TOEIC 940)